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Review article

A systematic review of the impact of the use of social networking sites on body image and disordered eating outcomes

Grace Holland*, Marika Tiggemann

School of Psychology, Flinders University, Adelaide, Australia

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ABSTRACT

A large body of literature has demonstrated mass media effects on body image and disordered eating. More recently, research in this area has turned to 'new' forms of media, such as the Internet, and particularly Social Networking Sites (SNSs). A systematic search for peer-reviewed articles on SNS use and body image and eating disorders resulted in 20 studies meeting specific inclusion criteria. As a whole, these articles demonstrated that use of SNSs is associated with body image and disordered eating. Specific SNS activities, such as viewing and uploading photos and seeking negative feedback via status updates, were identified as particularly problematic. A small number of studies also addressed underlying processes and found that appearance-based social comparison mediated the relationship between SNS use and body image and eating concerns. Gender was not found to be a moderating factor. It was concluded that, although there is a good deal of correlational research supporting the maladaptive effect of SNS use on body image and disordered eating, more longitudinal and experimental studies are needed.

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* Corresponding author at: School of Psychology, Flinders University, GPO Box 2100, Adelaide, South Australia 5001, Australia. Tel.: +61 434946683.
E-mail address: holl0210@uni.flinders.edu.au (G. Holland).

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Introduction

Research on the psychological implications of using and being exposed to social networking sites (SNSs) is a relatively novel area of research that is gaining momentum. To date, both beneficial and problematic implications of SNS use have been identified. For example, SNS use has been related not only to greater social connectedness and wellbeing (Valkenberg & Peter, 2009), but also to increased loneliness (Nie & Hillygus, 2002). An emerging literature has investigated the influence of SNSs on body image and disordered eating. The present paper aims to systematically review the available research in this area.

Body image can be defined as a person's perceptions, thoughts, and feelings about their body (Grogan, 2008). Body dissatisfaction occurs when views of the body are negative and involves a perceived discrepancy between a person's assessment of their actual and ideal body (Cash & Szymanski, 1995; Grogan, 2008). Research in both Australia and the United States suggests that women and adolescent girls experience higher levels of body dissatisfaction and disturbed eating patterns than do their male counterparts (Ata, Ludden, & Lally, 2007; Tiggemann, 2005). It is estimated that approximately 50% of adolescent girls report being unhappy with their bodies (Bearman, Presnell, & Martinez, 2006). This dissatisfaction can develop from as early as six years of age and research has shown that the issue exists amongst individuals of varying body shape and cultural background (Dohnt & Tiggemann, 2006; Grabe & Hyde, 2006). Further, it can lead to maladaptive consequences for both physical and mental health, including depression, anxiety, low self-esteem, and eating disorders (Dittmar, 2009; Grabe, Ward, & Hyde, 2008; Groesz, Levine, & Murnen, 2002).

Body dissatisfaction is generally attributed to social factors, with the mass media considered the most influential and pervasive cause (e.g., Thompson, Heinberg, Altabe, & Tantleff-Dunn, 1999; Tiggemann, 2011). Images of the 'ideal' body shape have infiltrated traditional mass media sources, such as woman's fashion magazines and television, for many years. These images have been found to promote unattainable beauty ideals of glamorous and often photo-shopped women who are generally young, tall, moderately breasted, and incredibly thin (Grabe et al., 2008; Tiggemann, 2011). The association between media exposure and body dissatisfaction and disordered eating among women and girls has been supported by extensive correlational (for meta-analyses, see Grabe et al., 2008; Levine & Murnen, 2009) and experimental research (for meta-analyses, see Grabe et al., 2008; Groesz et al., 2002; Want, 2009). A number of theories have attempted to explain the effect of the media on body image and eating behaviours, with two of the most commonly accepted frameworks being sociocultural and objectification theories.

According to sociocultural theory (Thompson et al., 1999; Tiggemann, 2011), the media present women with contemporary thin beauty ideals. Despite their impossibility, many women aspire to these ideals, but almost invariably fail to achieve them, resulting in body dissatisfaction (Ata et al., 2007; Keery, Van Den Berg, & Thompson, 2004; Stice, 1994; Tiggemann, 2002). Further, the theory proposes that the media's portrayal of unrealistic beauty ideals encourages women to both internalize them and engage in appearance comparisons with them. Thus, internalization and appearance comparison are highlighted as possible mechanisms in the development and maintenance of body dissatisfaction (Keery et al., 2004; Stice, 1994; Stice, Schupak-Neuberg, Shaw, & Stein, 1994). In support, a number of studies have shown that appearance comparison mediates the effects of traditional media on body image concerns (see meta-analysis by Groesz et al., 2002).

Another framework for understanding the relationship between the media and body image and disordered eating is provided by objectification theory (Fredrickson & Roberts, 1997). Essentially, objectification theory posits that in Western societies, the female body is socially constructed as an object to be looked at and evaluated, primarily on the basis of appearance. One example is the sexualized representation of women in visual media (both mainstream media and pornography). The pervasiveness and repeated occurrence of such sexual objectification leads to women and girls becoming acculturated to internalize an observer's perspective of their own bodies. That is, they come to perceive themselves as an object to be looked at and evaluated based on their appearance, a process termed 'self-objectification' (Fredrickson & Roberts, 1997). Self-objectification is a form of self-consciousness characterized by habitual and constant monitoring of the body's external appearance. Fredrickson and Roberts (1997) argue that this constant body surveillance leads to increases in both shame and anxiety about the body, which in turn contribute to a number of mental health issues including disordered eating. Experimental evidence has demonstrated that exposure to sexualized fashion magazine images results in both state self-objectification and body dissatisfaction (Harper & Tiggemann, 2008; Krawczyk & Thompson, 2015).

Although traditional media are still widely consumed, other forms of 'new' media are being increasingly accessed, most notably the Internet. Since 1990, Internet usage has increased by approximately 50% each year (Internet Growth Statistics, 2014). In July 2014, it was estimated that the Internet was used by 40.4% of the world's population (Internet Users, 2015). Moreover, image-focused websites (e.g., fashion/beauty and entertainment/celebrity websites) are currently more popular among undergraduate women than are fashion magazines (Bair, Kelly, Serdar, & Mazzeo, 2012). Recent studies demonstrate that, as in mainstream media, there exist many places on the Internet that perpetuate the stereotypical ideals of feminine beauty. For example, a content analysis of advertisements on websites aimed at adolescents found most figures in the advertisements were female, young, thin, and attractive (Slater, Tiggemann, Hawkins, & Werchon, 2011). In addition, studies of high school girls have found that Internet usage was related to greater internalization of thin ideal, appearance comparison, weight dissatisfaction, and drive for thinness (Tiggemann & Miller, 2010; Tiggemann & Slater, 2013). Similarly, Bair et al. (2012) found that young adult women who used the Internet were more likely to experience disordered eating.

One area of Internet usage that has become increasingly popular is the use of Social Networking Sites (SNSs). In January 2014, 74% of online adults were using SNSs (Social Networking Fact Sheets, 2015). These sites allow individuals to create public or private online profiles which they can then use to form relationships and interact with other users of the same website. Different from traditional mass media, a large proportion of SNSs are peer-generated, meaning that users are simultaneously information sources and receivers. In addition, compared to the passive consumption of traditional media forms, individuals actively decide how they participate on SNSs. For example, users have the ability to search for almost any other user or group they are interested in. They can also participate in a variety of activities, such as sharing photos, videos, and everyday information about their lives and making online comments about others' activities. Examples of SNSs include *Facebook*, *Instagram*, *Twitter*, and *MySpace*. *Facebook* is currently the most popular SNS with over 1.15 billion current users worldwide (Bullas, 2014). The present paper aims to systematically review the small but growing body of research that has investigated the influence of SNSs on body image. Specifically, it seeks to review studies investigating the general use of SNSs and their relationship to body image and disordered eating in unselected samples.

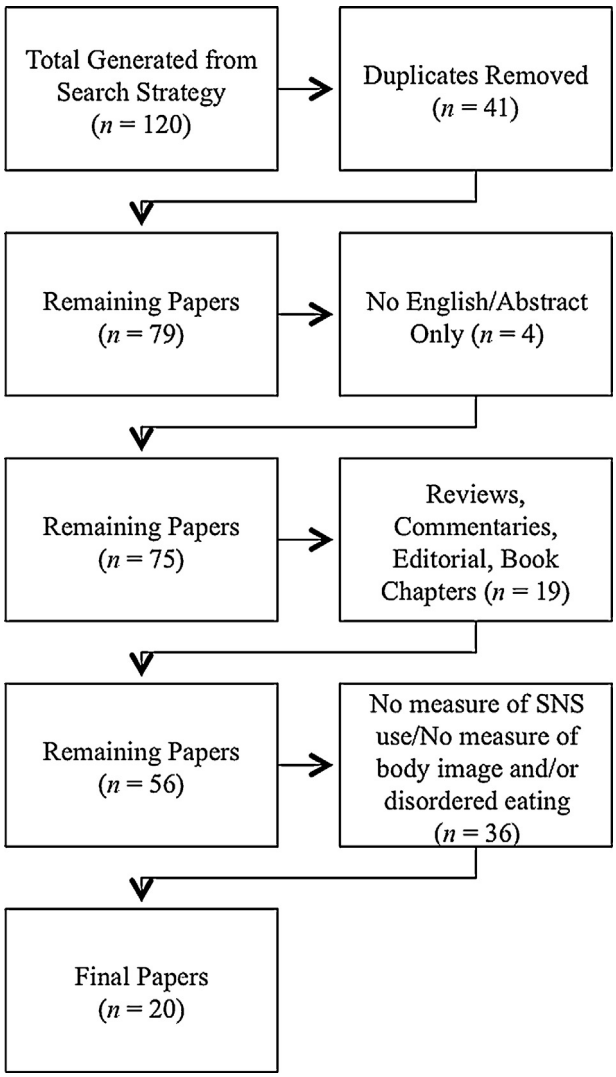

Fig. 1. Search strategy.

Method

Search Strategy

To identify papers for review, a search was performed in May 2015 using the PsychInfo and PubMed databases. Databases were searched using all combinations of the terms 'social media', 'social networking sites', 'facebook', 'twitter', 'instagram', 'tumblr', 'pinterest', and 'flickr', intersected with 'eating disorder*', 'disordered eating', 'body image', 'body dissatisfaction', 'drive for thinness', 'drive for muscularity', 'thin ideal', 'bulimia', 'anorexia', 'binge eating', and 'orthorexia'. References of the collected articles were also scanned for additional relevant studies.

Selection Criteria

The search strategy generated a total of 120 articles (see Fig. 1 for a diagrammatic representation of the search strategy). First, identified duplicates were removed ($n = 41$). Articles in languages other than English, published abstracts, reviews, commentaries, editorials, book chapters, and dissertations were also excluded ($n = 30$). To be included, articles needed to be peer-reviewed. Importantly, articles for this review needed to contain at least one measure of SNS use, as well as at least one measure of body image or disordered eating. These selection criteria led to the retention of 20 articles.

Results and Discussion

Selected Journal Articles

Of the 20 journal articles meeting all inclusion criteria, 16 involved cross-sectional studies, with five of these including a longitudinal component. One article had both a correlational and experimental study and the remaining three papers were experimental designs. Approximately half the studies had only female participants ($n = 11$), but nine included men, although they were always clearly the minority gender. Most studies had participants from the U.S. ($n = 11$), with two of these involving a comparison with Korean participants. The remaining studies were based on Australian ($n = 4$), Dutch ($n = 2$), English ($n = 1$), German ($n = 1$), and Belgian ($n = 1$) samples. See Table 1 for more information on the included articles.

Correlational Research

The earlier studies investigating the influence of SNSs on body image and disordered eating tended to be correlational. These studies aimed to investigate whether, as with traditional media, the use of SNSs relates to body image concerns and disordered eating. Several ways of measuring SNS use were used, including time spent on SNSs, the frequency with which they were checked and the number of online 'friends' users have. Some studies also explored specific activities and underlying mechanisms relevant to the use of certain SNSs.

Time on SNSs. The research in Table 1 demonstrates that the overall time spent on SNSs is related to indices of body image and disordered eating. Tiggemann and colleagues (Tiggemann & Miller, 2010; Tiggemann & Slater, 2013, 2014) found that spending more time on *MySpace* and *Facebook* was associated with increased body surveillance, greater endorsement of the thin ideal, more frequent appearance comparisons, and decreased weight satisfaction among younger girls and adolescent women. Similarly, adolescent women who spent more time on *Facebook* and more frequently checked their profiles, had higher drive for thinness and greater body dissatisfaction (Fardouly & Vartanian, 2014). Likewise, Mabe, Forney, and Keel (2014) reported a positive correlation between time spent on *Facebook* and disordered eating, whereas Vandenbosch and Eggermont (2012) found that the frequency with which people checked their SNS accounts was significantly related to greater levels of body surveillance. Finally, a study of the Dutch SNS *Hyves.nl* (comparable to *Facebook* and the most popular SNS among Dutch adolescents) revealed positive links between the frequency of checking this SNS and adolescents' appearance investment (De Vries, Peter, Nikken, & De Graaf, 2014).

Facebook "friends". As a measure of engagement with *Facebook*, some studies investigated the relationship between users' number of online 'friends' and body image and disordered eating. Rutledge, Gillmor, and Gillen (2013) found that having more *Facebook* 'friends' was beneficial as it was related to more positive views of appearance. However, the majority of studies suggested negative correlates of having a greater number of online 'friends'. Young girls with more *Facebook* friends reported higher levels of body surveillance, drive for thinness, internalization of beauty ideals, and dieting behaviours (Tiggemann & Slater, 2013, 2014). Likewise, a recent study found that having more *Facebook* friends was correlated with both drive for thinness and appearance comparison in undergraduate students (Kim & Chock, 2015). The difference

Table 1
Summary of articles in this review.

Author (year)	Country	Design[a]	Sample (*N*)	%Female	SNS measure	Body image/eating disorder measure	Key findings	Risk of bias
De Vries et al. (2016)	Netherlands	C L	Adolescents (604) (Community)	51	Frequency of checking Hyves.nl each day	Body dissatisfaction Peer appearance-related feedback	SNS use predicted increased body dissatisfaction SNS use predicted increased peer influence on body image via peer appearance-related feedback Gender did not moderate findings	Unclear
De Vries et al. (2014)	Netherlands	C L	Adolescents (604) (Community)	51	Frequency of checking Hyves.nl each day	Appearance investment Desire to undergo cosmetic surgery	SNS use predicted desire to undergo cosmetic surgery indirectly via increased appearance investment Gender did not moderate findings	Unclear
Fardouly et al. (2015)	UK	E	University students & staff (112)	100	10 min on own Facebook account vs control website	State appearance-discrepancy State body dissatisfaction Trait appearance comparison tendency	Greater appearance comparison tendency related to greater facial, hair, and skin-related discrepancies after Facebook exposure	Low
Fardouly and Vartanian (2014)	Australia	C	Undergraduates (227)	100	Frequency of checking Facebook each day Length of time on Facebook per day	Body dissatisfaction Drive for thinness General appearance comparisons Specific appearance comparisons (close friends/family/distant peers/celebrities)	Facebook use related to body image concerns Results mediated by appearance comparisons in general, frequency of comparisons to close friends and distant peers, and by upward comparisons to distant peers and celebrities	Unclear
Ferguson et al. (2012)	USA (2.5% White)	C L	Pre-adolescents and adolescents (Community) (237)	100	Frequency of checking various SNSs per day	Body dissatisfaction Eating disorder symptoms Peer competition	SNS use not related to body dissatisfaction or eating disorders SNS use contributed to later peer competition	Unclear
Haferkamp and Kramer (2011)	Germany	E	Young adults (Community) (91)	53	Facebook profiles of attractive vs unattractive users	Body satisfaction	Viewing more attractive users lead to less body satisfaction	Low
Hummel and Smith (2014)	USA (73.2% White)	C L	Undergraduates (185)	78	Facebook feedback seeking Facebook status & comment coding	Eating disorder symptoms	Negative feedback seeking style and receiving a high number of Facebook comments was related to greater likelihood of later disordered eating Receiving negative comments in response to personal information in status updates was related to greater likelihood of later disordered eating	Unclear

Table 1 (*Continued*)

Author (year)	Country	Design[a]	Sample (*N*)	%Female	SNS measure	Body image/eating disorder measure	Key findings	Risk of bias
Kim and Chock (2015)	USA (74.1% White)	C	Undergraduates (186)	64	Length of time on Facebook per day Number of Facebook friends Amount of social grooming behaviours on Facebook	Appearance comparison Drive for muscularity Drive for thinness	Time spent on SNS was not related to body image concerns. Number of friends related to appearance comparison and drive for thinness Social grooming correlated with drive for thinness mediated by appearance comparison	Unclear
H.R. Lee et al. (2014)	USA (20.8% White)	C	Undergraduates (1020)	USA: 60 Korea: 63	Types of information sought when using SNSs	Body satisfaction	SNS use for information about body image negatively related to body satisfaction in US & Korea SNS use for self-status seeking regarding body image related to body satisfaction in Korea	Unclear
H.E. Lee et al. (2013)	USA (29% White)	E	Undergraduates (296)	100	Underweight Facebook profile picture vs overweight Facebook profile picture	Body satisfaction	Koreans who witnessed an underweight peer talk negatively about their body reported lower body satisfaction than those witnessing an overweight peer talk negatively about their body	Low
Mabe et al. (2014)	USA (77.4% White)	C E	Undergraduates (1044)	100	Length of time on Facebook per week 20 min on own Facebook account vs control websites	Eating disorder symptoms	Facebook use related to greater disordered eating Facebook use related to the maintenance of weight/shape concerns	Unclear – C Low – E
Manago et al. (2015)	USA (73.8% White)	C	Undergraduates (815)	57	Length of time on Facebook per day Investment in Facebook Passive Facebook use Active Facebook use	Appearance self-worth Body shame Body surveillance	Facebook use predicted objectified body consciousness, which in turn predicted greater body shame	Unclear
Meier and Gray (2014)	USA (84.5% White)	C	Adolescents (Community) (103)	100	Length of time on Facebook per day Frequency of Facebook appearance-related exposure	Appearance comparisons Drive for thinness Internalization of the thin ideal Self-objectification Weight satisfaction	Facebook use not correlated with body image or disordered eating concerns Greater appearance-exposure correlated with weight dissatisfaction, drive for thinness, thin ideal internalization, and self-objectification	Unclear
Rutledge et al. (2013)	USA (43% White)	C	Undergraduates (225)	54	Length of time on Facebook per week Number of Facebook friends Emotional investment in Facebook	Appearance evaluation Appearance orientation	Emotionally invested in Facebook related to appearance-orientation More Facebook friends related to more positive views of appearance	Unclear

Table 1 (*Continued*)

Author (year)	Country	Design[a]	Sample (*N*)	%Female	SNS measure	Body image/eating disorder measure	Key findings	Risk of bias
Smith et al. (2013)	USA (76.3% White)	C L	Undergraduates (232)	100	Maladaptive Facebook usage	Body dissatisfaction Bulimic symptoms Over-eating Shape concern	Maladaptive Facebook usage predicted increases in bulimic symptoms and episodes of over-eating Body dissatisfaction mediated the relationship between maladaptive Facebook use and increases in over-eating Body dissatisfaction partially mediated relationships between maladaptive Facebook use and bulimic symptoms.	Unclear
Thompson and Lougheed (2012)	USA (76.5% White)	C	Undergraduates (268)	76	Viewing Facebook pictures	Negative body image	Females more likely than males to report that Facebook pictures caused negative body image	Unclear
Tiggemann and Miller (2010)	Australia	C	High school students (156)	100	Length of time on SNSs per day Frequency of checking SNS each day	Appearance comparison Drive for thinness Internalization of the thin ideal Weight satisfaction	Facebook and MySpace usage positively correlated with endorsement of the thin ideal, appearance comparison and decreased weight satisfaction	Unclear
Tiggemann and Slater (2013)	Australia	C	High school students (1087)	100	Length of time on SNSs per day Number of friends on SNSs	Body surveillance Drive for thinness Internalization of beauty ideals	Facebook users scored significantly more highly on all body image concern measures than non-users. Number of Facebook friends was significantly correlated with all body image concern measures.	Unclear
Tiggemann and Slater (2014)	Australia	C	Primary school students (189)	100	Length of time on SNSs per day Number of friends on SNSs	Body esteem Body surveillance Dieting Internalization of the thin ideal	SNS use correlated with internalization, body surveillance, negative body esteem, and dieting Number of Facebook friends significantly correlated with dieting	Unclear
Vandenbosch and Eggermont (2012)	Belgium	C	High school students (558)	100	Frequency of visiting SNS per day	Body surveillance Internalization of beauty ideals Self objectification	Direct relationships between internalization of beauty ideals and sexually objectifying media Indirect relationships between self-objectification and body surveillance with sexually objectifying media through the internalization of beauty ideals SNS use was the only medium that significantly predicted body surveillance	Unclear

[a] C, correlational; L, longitudinal; E, experimental.

from the earlier results may be explained by the fact that the latter two studies had a larger proportion of female participants than the study by Rutledge et al. (2013), although it should be noted that the studies did not explicitly assess outcomes across gender.

Specific Facebook activities. As SNSs encourage active use over passive consumption, the overall time spent on them might not satisfactorily capture the range of interactive elements existing on these sites. Therefore, some studies have attempted to measure Facebook use as multi-faceted. For example, Manago, Ward, Lemm, Reed, and Seabrook (2015) measured *Facebook* 'involvement' as the combination of time and investment on the site as well as the amount of time spent actively and passively using it. Their combined measure was associated with objectified body consciousness, which in turn predicted greater body shame (Manago et al., 2015).

Other studies have moved away from measures of overall *Facebook* use and have concentrated on specific ways in which users interact with it. For example, despite finding no connection with overall time spent on the SNS, Rutledge et al. (2013) found that individuals who felt more positively towards *Facebook* were also more appearance-oriented. In another study, Smith, Hames, and Joiner (2013) measured the relationship between maladaptive *Facebook* use (the tendency to seek negative evaluations and engage in social comparisons) and eating pathologies. According to their results, maladaptive *Facebook* use was related to greater levels of bulimic symptoms, body dissatisfaction, and shape concerns (Smith et al., 2013). Similar results were found by Hummel and Smith (2014) who observed actual *Facebook* status updates. They found that users who wrote statuses with a negative feedback seeking style were more likely to report shape, eating, and weight concerns (Hummel & Smith, 2014).

In a detailed examination of *Facebook* components, Meier and Gray (2014) determined that overall time on *Facebook* was not related to body image concerns, but that engaging in photo-based activities (e.g., posting and sharing photos of oneself and friends) was. More specifically, Meier and Gray (2014) found that sharing more photos on *Facebook* led to a greater likelihood of basing self-worth on appearance and a higher endorsement of the thin ideal. Moreover, viewing other users' photos was positively correlated with weight dissatisfaction, drive for thinness, thin ideal internalization, and self-objectification (Meier & Gray, 2014). Similarly, Kim and Chock (2015) found that 'social grooming' behaviours (e.g., viewing, 'liking', and making comments on other *Facebook* users' photos and statuses) was related to higher drive for thinness and appearance comparison.

In summary, recent studies have moved away from focusing on overall SNS use. Instead, they have examined more specific aspects of SNS use and have found that these components provide clearer information about how SNS use is related to body image and eating concerns. Findings suggest that photo-based activities are particularly salient.

Underlying processes. Sociocultural theory (Thompson et al., 1999) suggests that women and girls internalize the media's portrayal of unrealistic beauty ideals and engage in appearance comparisons with them. Similarly, objectification theory (Fredrickson & Roberts, 1997) holds that the constant promotion of (often sexualized) beauty ideals in the media lead women and girls to self-objectification and constant body surveillance. All these processes (internalization, appearance comparisons, self-objectification, and body surveillance) are argued to underlie increases in body dissatisfaction and disordered eating.

In the context of SNSs, Fardouly and Vartanian (2014) examined how online appearance comparisons to *Facebook* images of family members, close friends, distant peers, and celebrities affected body image. Their results showed that the frequency of comparisons with close friends and distant peers, and upward comparisons with close friends, distant peers, and celebrities, mediated the relationship between time spent using *Facebook* and body image concerns. Further, Vandenbosch and Eggermont (2012) found that internalization of beauty ideals accounted for the relationship between viewing sexually objectified images on SNSs and both self-objectification and body surveillance. Similarly, Manago et al. (2015) found that Facebook use was related to greater objectified body consciousness, which was in turn found to predict greater body shame. In accord with sociocultural and objectification theories, these results suggest that spending more time on *Facebook* is linked to body dissatisfaction among women at least partly due to the internalization of unrealistic ideals, self-objectification, appearance comparison, and self-surveillance.

Longitudinal Designs

The aforementioned cross-sectional studies have demonstrated a relationship between SNS use and body image and disordered eating, but are limited in their ability to determine directionality of effects. Thus, some studies have employed a longitudinal research design in order to determine the temporal sequencing of variables.

Both Smith et al. (2013) and Hummel and Smith (2014) reassessed a subset of participants in their correlational study after a period of four weeks. Their results showed that using *Facebook* in ways that invited negative feedback and social comparison predicted greater levels of body image concerns and disordered eating four weeks later. Hummel and Smith (2014) also found that individuals who had written more personally revealing status updates on *Facebook* on which they received negatively tinged comments, were more likely to have subsequent increased levels of eating, weight, and shape concern.

Ferguson, Munoz, Garza, and Galindo (2012) measured the influence of SNS use and peer competition on body satisfaction and eating disorder symptoms among teenage girls with a six-month follow-up period. They found no concurrent or prospective correlation between SNS use and body dissatisfaction or eating disorder symptoms. This lack of relationship may have been due to their measure of SNS use, which included activities like online gaming and blogging, or the fact that the majority of participants were Latino (94.1%). Nevertheless, SNS use was found to contribute to higher levels of peer competition six months later, and peer competition itself was related to subsequent body dissatisfaction and disordered eating (Ferguson et al., 2012). Thus it appears SNS use had an indirect impact on body dissatisfaction and disordered eating.

The most recent longitudinal studies involved users of the Dutch SNS *Hyves.nl* over an 18-month period (De Vries, Peter, De Graaf, & Nikken, 2016; De Vries et al., 2014). De Vries et al. (2014) found that more frequent use of *Hyves.nl* predicted a subsequent increase in appearance investment among adolescents, which was in turn related to an increased desire for cosmetic surgery. Likewise, De Vries et al. (2016) showed that more frequent SNS use predicted increased body dissatisfaction 18 months later. Importantly, in both these studies, whereas more frequent SNS use predicted subsequent body dissatisfaction, initial body dissatisfaction did not predict subsequent SNS use. Thus it was concluded that SNS use is temporally antecedent to body image concerns, consistent with a postulated causal role.

Collectively, the majority of these longitudinal studies provide preliminary evidence that SNS use is a predictor of subsequent body image and eating concerns up to 18 months later. This suggests that SNS use may indeed play a causal role in the development of body dissatisfaction and disordered eating.

Experimental Research

Although correlational research to date has established significant associations between SNS use and body dissatisfaction and eating pathology, and a small amount of longitudinal research has suggested temporal sequencing of this relationship, only experimental studies can actually determine causality. As yet, there are only four experimental studies that have investigated the effect of SNS use on body image and eating concerns.

To replicate studies on traditional media showing that exposure to beauty ideals can lead to negative consequences in terms of body image and disordered eating, two experimental studies presented participants with *Facebook* profiles of users with different physical characteristics. Specifically, Haferkamp and Kramer (2011) presented participants with *Facebook* profiles of attractive or unattractive users. Participants reported feeling greater body dissatisfaction immediately after observing more attractive users than when they were exposed to less attractive profile pictures (Haferkamp & Kramer, 2011). In similar fashion, Lee, Taniguchi, Modica, and Park (2013) presented their participants with profile pictures of underweight or overweight users on *Facebook*. They found that Korean undergraduates who witnessed an underweight peer make online comments about a desire to lose weight, demonstrated lower body satisfaction than those who witnessed an overweight peer's desire to lose weight (H.E. Lee et al., 2013).

Two later experimental studies considered how the influence of SNS use on body image and disordered eating might differ from that of traditional media forms. First, Mabe et al. (2014) used an experimental design to compare the influence of spending 20 min on *Facebook* with alternative websites. They found that women in the *Facebook* condition exhibited greater concern about their weight and shape than women in the alternative website condition (Mabe et al., 2014). Further, effects in Mabe et al.'s study were noted after only 20 min exposure to *Facebook*, whereas statistics suggest that the average Internet user logs approximately 1.72 h per day on SNSs (Mander, 2015). Hence the effects demonstrated in this research may be even larger when generalized to real-world settings.

Second, Fardouly, Diedrichs, Vartanian, and Halliwell (2015) compared the effect of spending 10 min browsing *Facebook* as opposed to a fashion magazine website or an appearance-neutral control website. In contrast to Mabe et al.'s (2014) result, Fardouly et al. (2015) found that the *Facebook* condition had no effect on participants' body satisfaction nor their desire to change their weight and shape (Fardouly et al., 2015). However, women higher in appearance comparison tendency did report more facial, hair, and skin related concerns in the *Facebook* condition (Fardouly et al., 2015). This perhaps reflects the fact that profile pictures on *Facebook* are often upper-body or headshots. Another explanation may be that 10 min exposure is not sufficient to affect body satisfaction, in contrast to Mabe et al.'s (2014) finding with 20 min of exposure.

Gender

As is the case with research on traditional media, most studies of the impact of SNS on body image and disordered eating have been conducted using female samples. However, as men and adolescent boys also experience body dissatisfaction and eating disorder symptoms (McCabe & Ricciardelli, 2001; Striegel-Moore et al., 2009) and are frequent users of SNSs (Social Networking Fact Sheets, 2015), six articles in this review specifically examined gender differences. Of these, only one study found a significant difference in relationships between genders. Thompson and Lougheed (2012) showed that significantly more female *Facebook* users agreed that photos on *Facebook* caused them to experience body dissatisfaction than did male users. However, the finding was based on only one questionnaire item.

All other studies that investigated gender differences in the relationship between SNSs and body image or disordered eating found no significant differences. Haferkamp and Kramer (2011) found both men and women had a decrease in body satisfaction after being exposed to attractive *Facebook* profiles. Similarly, De Vries et al. (2014) found that SNS use was associated with greater appearance investment and desire for cosmetic surgery across both genders with no significant differences. Parallel to this, De Vries et al. (2016) found that SNS use was related to men and women's body image in the same manner and to the same extent. Likewise, Manago et al. (2015) found that *Facebook* involvement was a significant predictor of objectified body consciousness and greater body shame across gender. Finally, Kim and Chock (2015) found that both female and male participants with a greater tendency to engage in online 'social grooming' behaviours were more likely to have a higher drive for thinness.

Culture

Research generally suggests that the influence of traditional media on body image and disordered eating is increasingly similar across cultures (Grabe & Hyde, 2006). In terms of SNSs, as yet, there have only been two cross-cultural studies. Both of them compared students attending a Korean university with U.S. students from a Hawaiian university.

Lee, Lee, Choi, Kim, and Han (2014) measured socializing, information, and self-status seeking needs (the desire to seek personal status via SNSs for career development and in order to feel influential) as functions of SNS use and found similarities and one difference between American and Korean participants. Both Korean and American participants who were higher in information seeking (the desire to learn about events and specific products and services) were more likely to have a negative body image (H.R. Lee et al., 2014). The only significant difference found was that Korean participants who used SNSs for self-status seeking purposes had more positive body satisfaction, whereas this relationship was not found for American users (H.R. Lee et al., 2014).

In the other study, H.E. Lee et al. (2013) found that only Korean participants reported lower body satisfaction after being exposed to pictures of underweight peers on *Facebook* compared to overweight peers. Peer body size did not affect the American participants (H.E. Lee et al., 2013). However, the stimuli used were photographs of an Asian woman and thus, as social-comparison theory (Festinger, 1954) suggests, the Korean participants may have been more likely to compare themselves with the profile photos than the U.S. participants.

The differences noted in the above studies between the U.S. and Korean participants suggest that the way in which users seek information on SNSs may vary depending on their culture. However, the current cross-cultural research involves only Korean and U.S. samples, and thus other samples need to be investigated. Nevertheless, the studies by H.E. Lee et al. (2013) and H.R. Lee et al. (2014) show that SNS use can have damaging implications in non-western samples.

Quality of Evidence

The Cochrane Collaboration's tool for assessing risk of bias (Higgins et al., 2011) was employed to assess the risk of bias in the 20 studies included in the present review (see last column in Table 1). This tool assesses bias using seven domains: sequence generation, allocation concealment, blinding of participants, personnel and outcome assessors, incomplete outcome data, selective outcome reporting, and other sources of bias.

According to the Cochrane Collaboration tool (Higgins et al., 2011), a study is considered at low risk of bias when all domains

are labelled as such. The first four domains of the Cochrane Collaboration tool are most applicable to experimental research designs. Following this protocol, the four experimental studies included in the present review (Fardouly et al., 2015; Haferkamp & Kramer, 2011; H.E. Lee et al., 2013; Mabe et al., 2014) were assessed as having low risk of bias. Each of the five longitudinal studies was assessed as low risk for the domains of incomplete outcome data and selective outcome reporting. However, it was unclear as to whether bias was present in the other five domains and hence all longitudinal studies were given an overall rating of unclear. Similarly, although there was no specific indication of bias, the correlational studies reviewed were assessed as having an unclear risk of bias because they were all convenience samples. Nevertheless, it should be noted that, despite the majority being assessed as having an unclear risk of bias, all the studies reviewed provided clear detail in their procedures and discussed methodological limitations appropriately.

Conclusions and Implications

Although a novel topic of research, interest in the relationship between SNS and body image and disordered eating is growing rapidly. Hence the present article aimed to systematically review research that has examined this relationship. Results from 20 peer-reviewed journal articles provided evidence across different methodologies that SNS use has been associated with increased body dissatisfaction and disordered eating. Furthermore, these results held across gender.

Earlier correlational studies included in the review found that different measures of overall SNS use were related to body image concerns and disordered eating behaviour. These measures included overall time spent on SNSs (Fardouly & Vartanian, 2014; Mabe et al., 2014; Tiggemann & Miller, 2010; Tiggemann & Slater, 2013, 2014), the frequency of SNS use (De Vries et al., 2016, 2014; Vandenbosch & Eggermont, 2012), and the number of *Facebook* 'friends' (Kim & Chock, 2015; Tiggemann & Slater, 2013, 2014). Furthermore, longitudinal research demonstrated that overall SNS use, as indicated by these measures, resulted in increased levels of body image concerns between 4 weeks and eighteen months later (De Vries et al., 2016, 2014; Hummel & Smith, 2014; Smith et al., 2013).

Despite these results, other studies suggested that measures of overall SNS use did not properly capture the active nature of these sites. Hence, some research examined unique SNS features and found that greater emotional investment and negative feedback seeking were related to body image and eating concerns. Most salient are findings that showed how engagement in photo-based activities on *Facebook* was related to body dissatisfaction and disordered eating (Meier & Gray, 2014). Thus, future research would benefit from investigating more image-based SNSs, such as *Instagram*. In addition, *Instagram* provides an in-built tool that offers a number of possible filters to enhance the appearance of a photo. The use of such filters or other digital alteration techniques represents another important photo-based activity that warrants future research attention.

Some studies included in this review found that the underlying processes of internalization and appearance comparison accounted for the observed maladaptive effects of SNS use (Fardouly et al., 2015; Vandenbosch & Eggermont, 2012), in accordance with sociocultural theory (Festinger, 1954; Tiggemann, 2011). Indeed, it has been suggested that social comparison may be particularly pertinent to SNSs (Perloff, 2014). First, the speed and ease with which individuals can connect to their peers on SNSs provides greater opportunity for effortless and frequent social comparisons (Tiggemann & Miller, 2010). Second, in accordance with social comparison theory (Festinger, 1954), the drive for self-evaluation causes people to seek out comparisons with similar rather than dissimilar others. Thus peers provide more important appearance-comparison targets than models or celebrities (Heinberg & Thompson, 1995). As SNSs are largely peer-based, there is a greater potential for social comparison to attractive peers and subsequent negative views of oneself (Perloff, 2014). Other studies showed that the underlying processes of self-objectification and body surveillance also played a role in the relationship between SNS use and body dissatisfaction (Manago et al., 2015; Vandenbosch & Eggermont, 2012), in accordance with objectification theory (Fredrickson & Roberts, 1997). Just like traditional media, SNSs are a likely platform for objectified images of women. Future studies might usefully consider other processes proposed by other theoretical frameworks.

The findings of this systematic review point to a number of practical implications. The most obvious is for recommendations that individuals limit their exposure to SNSs. However, such a recommendation would be almost impossible to implement as social networking has become an increasingly common means of general communication for individuals in contemporary society. Instead, greater education around the processes and potential consequences involved in social networking might prove more effective. In particular, media literacy programmes, which have shown some success in combating negative body image (Levine & Murnen, 2009), now need to be expanded to incorporate consideration of social networking media, especially the pressures associated with the uploading and viewing of images.

As in all research, individual studies in this review need to be considered in the context of their limitations. However, as a whole body of research, a few particular limitations stand out. First, participant characteristics were not representative of the global community. Most studies involved undergraduate or high school students. This is not surprising as these age groups extensively use SNSs and are prone to body image and disordered eating issues. Nevertheless, both younger and older aged samples should be investigated. Currently, there is only one study with preadolescent girls (Tiggemann & Slater, 2014) which showed that young girls who use SNSs also have body image concerns. To date, there are no studies that have investigated the impact of SNS use on the body image and eating concerns of middle-aged or older women. Yet older women also experience body dissatisfaction (e.g., Clarke, 2002; Hetherington & Burnett, 1994) and are judged in relation to an ideal of female beauty that emphasizes both youthfulness and thinness (Tunaley, Walsh, & Nicolson, 1999). Moreover, the majority of mature adults (79% of 30–49 year-olds; 60% of 50–64 year-olds) use *Facebook* (Social Networking Fact Sheets, 2015). In addition, most of the studies investigated samples of White women and so research with ethnic minority groups is necessary. Furthermore, there are only two existing cross-cultural studies that found both similarities and differences between Korean and U.S. samples (H.E. Lee et al., 2013; H.R. Lee et al., 2014). Thus, future research will need to clarify the role played by culture in the relationship between SNS use and body image or eating concerns.

Second, there are a variety of methodological limitations in the research to date. For example, the majority of studies in this review investigated *Facebook*, as it is currently the most popular SNS. However, many other SNSs exist and it is worth noting that 42% of U.S. adults have reported frequenting more than one site (Duggan & Smith, 2013). Future research would benefit from investigating and comparing the influence of other SNSs in addition to *Facebook*. Another methodological limitation is that measures of SNS use usually consisted of only one or two items that varied between studies. A consistent definition and associated reliable measure of SNS use should be developed so that further studies can be validly replicated and compared. In addition, no current studies have taken into account the different technological platforms from which SNSs are

accessed. For example, there are currently 71% of SNS users accessing the sites from mobile devices (Bullas, 2014). Further studies should take into account the fact that access to SNSs via mobile devices is easier and more constantly available than access via desktop computers. Finally, the majority of studies reviewed investigated a range of body image variables and a minority investigated disordered eating. Future research is needed to obtain clarity on these different dependent variables.

Another major limitation lies in the research design of the studies to date. Although all the experimental studies in this review were assessed as having a low risk of bias according to the Cochrane Collaboration's tool (Higgins et al., 2011), the remaining studies were assessed as having an unclear risk (generally due to the use of convenience sampling). Future research in this area might address ways to counter potential bias caused by limited sampling techniques. Relatedly, the body of work on SNS use and body image and disordered eating is largely based on correlational findings. As yet, there are only a small number of longitudinal and experimental research designs. Hence, it is difficult to draw firm causal conclusions concerning the relationship between SNS use and body image and eating concerns. More longitudinal research that tracks both SNS use and body image and eating concerns over some considerable period of time, as well as more experimental research that manipulates exposure to SNSs, are needed to determine causal impact.

Finally, the current systematic review is not without its own limitations. As this field of research is constantly growing, there are likely studies that have been published or are in the process of submission that have not been included in the review. Further, the review focused on studies that investigated unselected SNSs, rather than websites dedicated to specific content (e.g., pro-anorexia websites). In addition because of the variety of measures employed in the studies, the review did not distinguish between specific outcome variables and this may have implications for the conclusions drawn.

In conclusion, the present systematic review has offered an important distillation of the current state of knowledge on the relationship between SNS use and body image and disordered eating. Across a number of measures and methodologies, the research has demonstrated that SNS use is significantly linked to body image and eating concerns. As the use of SNSs has become an integral part of contemporary society, it is important that future research continue to clarify the role played by SNS use in body image and disordered eating.

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